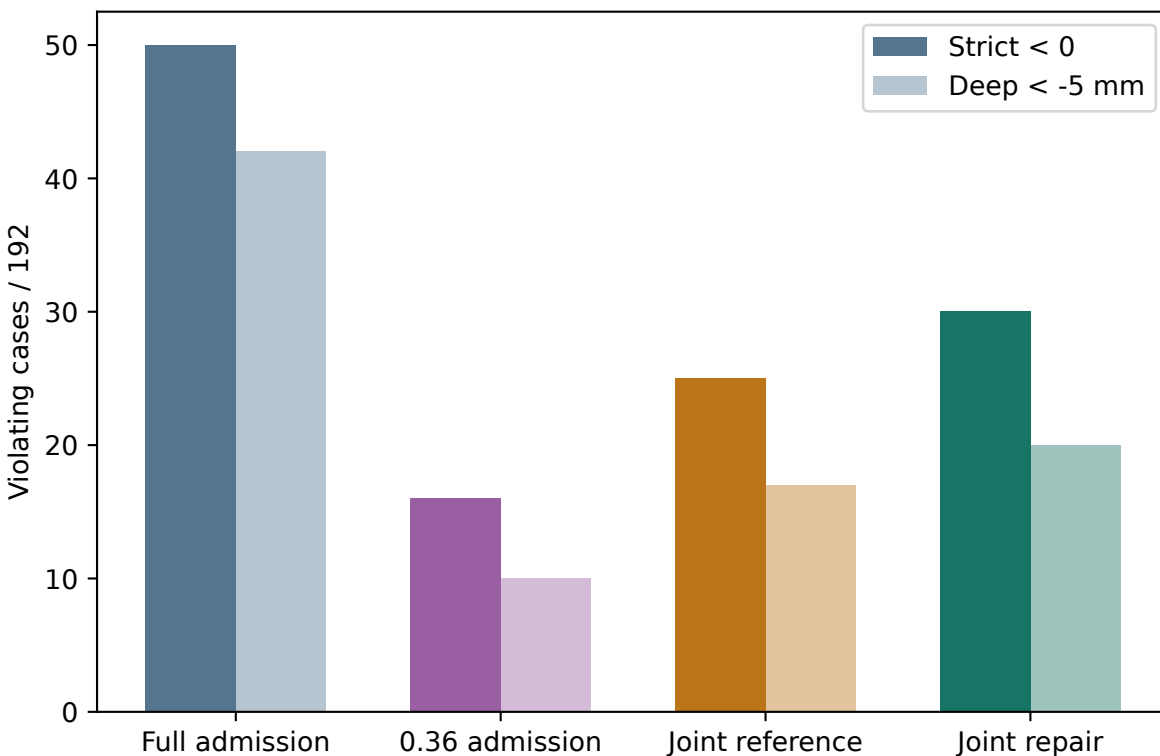
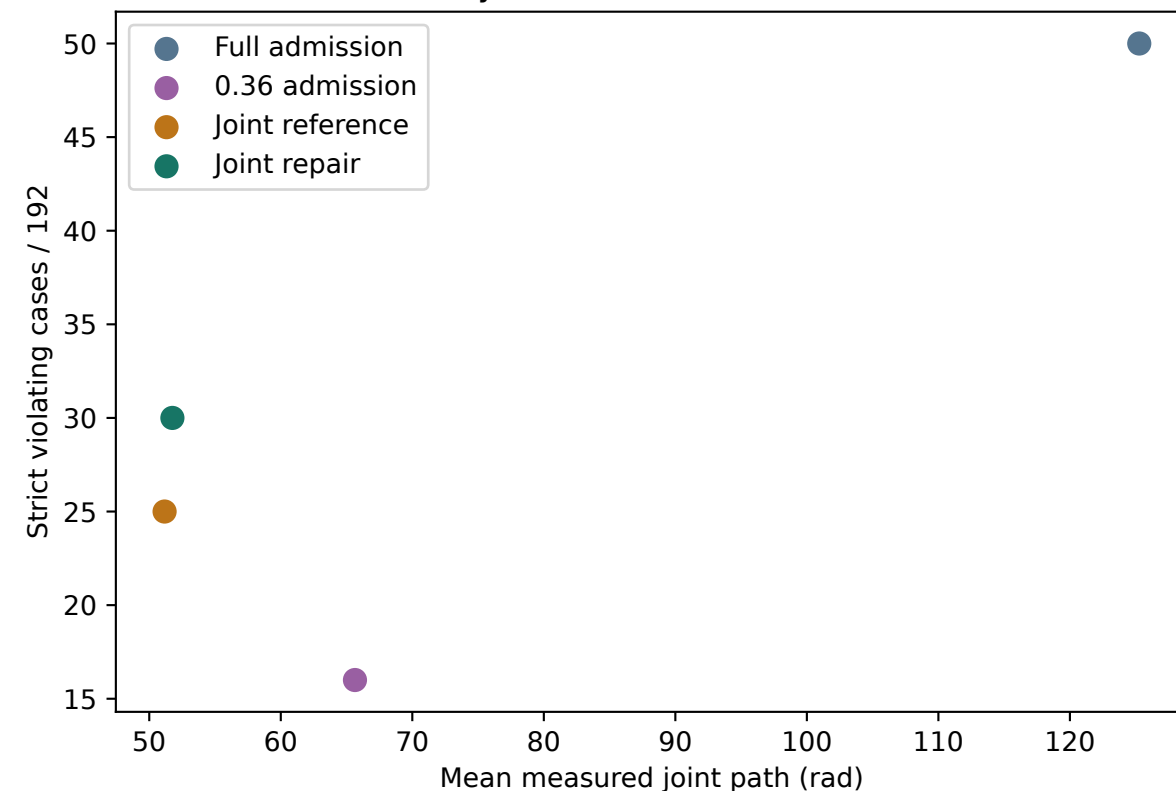


SafeDuo: 192 fresh conditioned paired cases; four frozen methods
Simulation evidence only; no hardware or full-space reliability claim

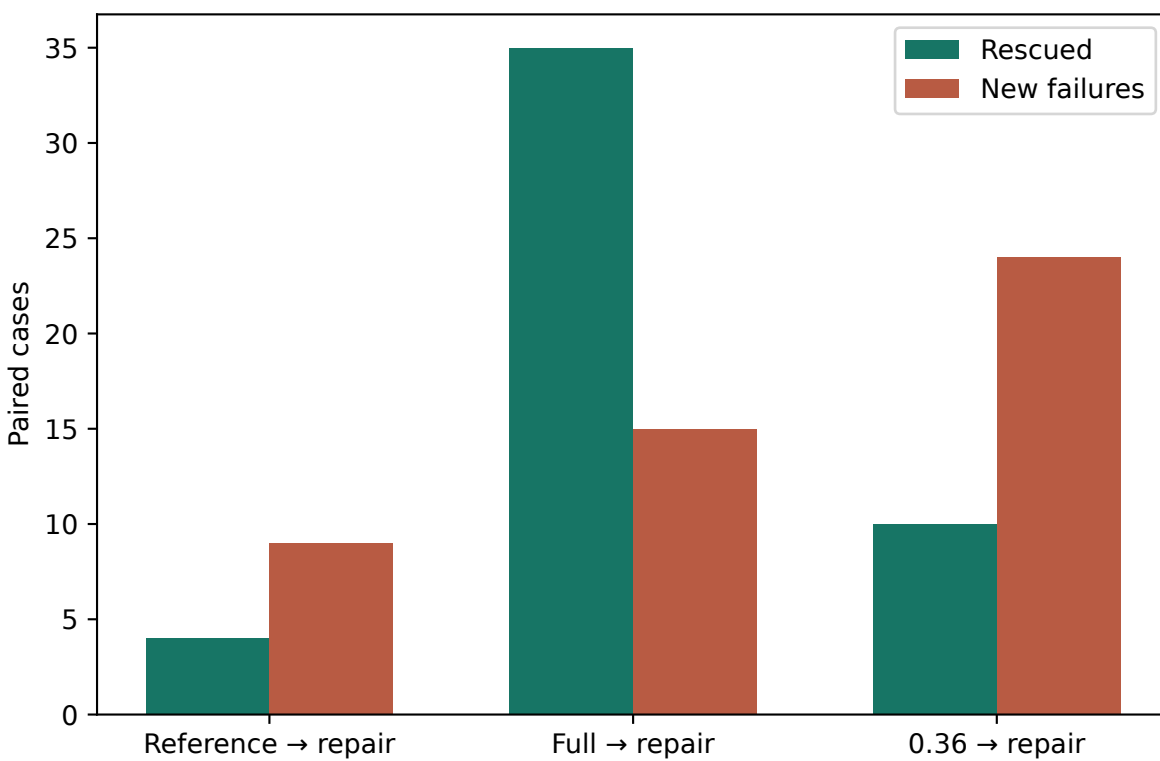
Observed failures: all 960 frames



Safety and delivered movement



Failures introduced remain in the denominator



Actual assigned-pair exposure (≥ 3 frames < 80 mm)

